

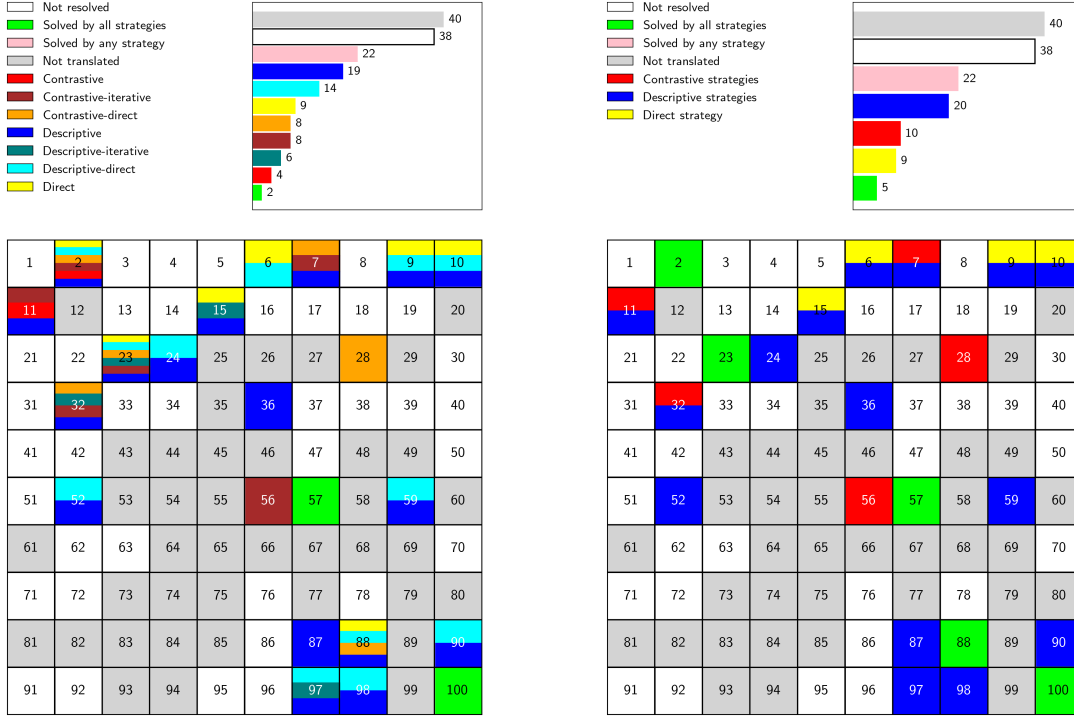


Figure 22: **Overall result of each strategy on Bongard-RWR.** Colormaps depict all problems solved by any tested model using the respective prompting strategy. The right figure aggregates strategies into corresponding groups for better coverage exposure.

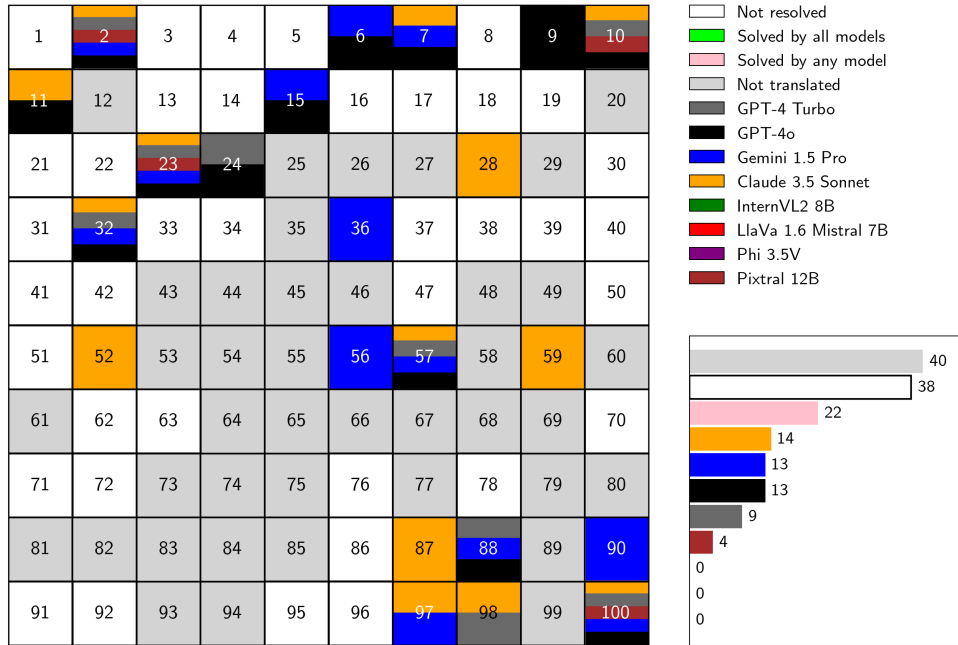
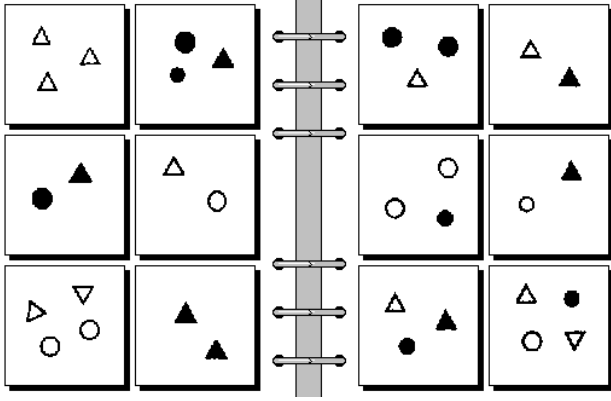
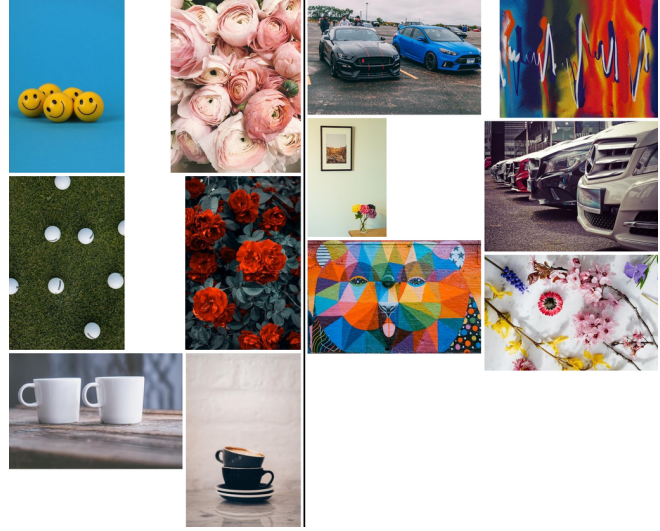


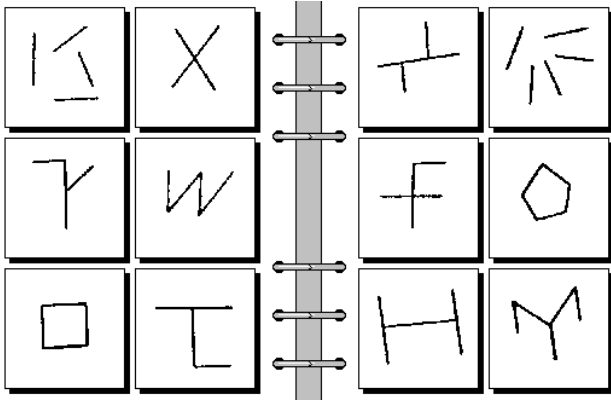
Figure 23: **Final summary of all experiments on the Bongard-RWR dataset.** The colormap aggregates all problems solved by any tested model using any generation prompting strategy. Overall, the models collectively managed to solve 22 problems.



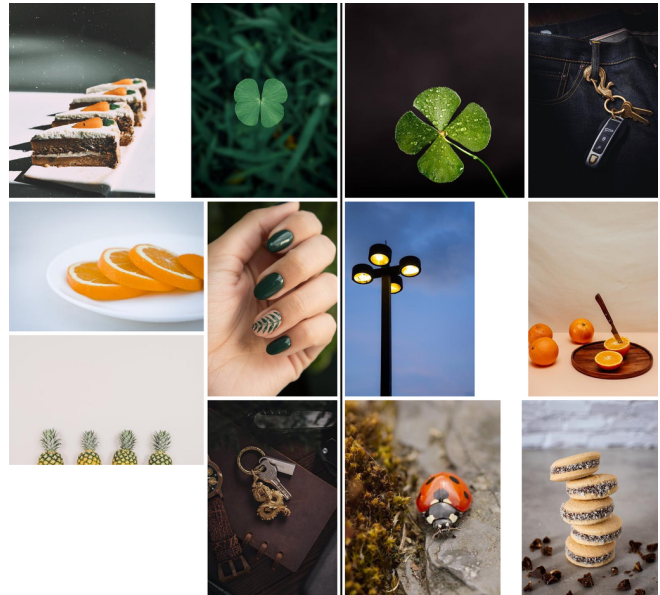
(a) Model's answer: "Left: All shapes are filled. Right: At least one shape is unfilled". Evaluated as incorrect.



(b) Model's answer: "Left: Images are monochromatic (containing only shades of a single color). Right: All images contain at least one hollow (unfilled) shape". Evaluated as correct.

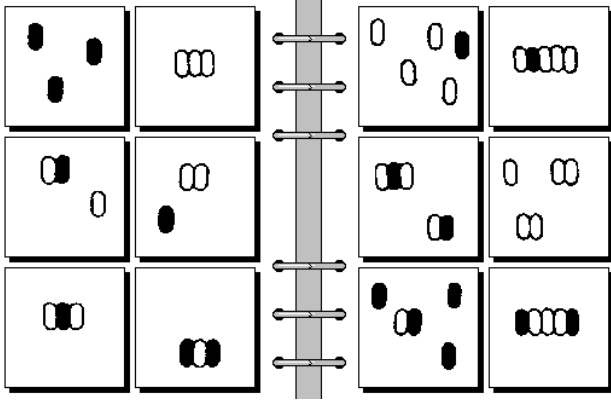


(a) Model's answer: "Left: All shapes are composed of straight lines that form only acute or right angles. Right: All shapes contain at least one instance of obtuse angles or curves (e.g., pentagon, 'H' shape, radial pattern)". Evaluated as incorrect.



(b) Model's answer: "Left: All images contain exactly four of something: four orange slices, four cake slices, four lobes of the four-leaf clover, four pineapples, four fingers with painted nails (thumb not visible), four visible elements (key, keychain, watch/compass, leather strap). Right: Images do not contain exactly four of any element: single four-leaf clover, multiple keys on a ring, single street lamp with multiple lights, single ladybug, multiple oranges (more than four), stack of five cookies". Evaluated as correct.

Figure 25: Synthetic BP #87 with its manually translated Bongard-RWR version. Correct answer: "Left: Four parts. Right: Five parts". Provided answers belong to Claude 3.5 Sonnet using the Descriptive strategy, as it was the only combination that solved this Bongard-RWR instance correctly.

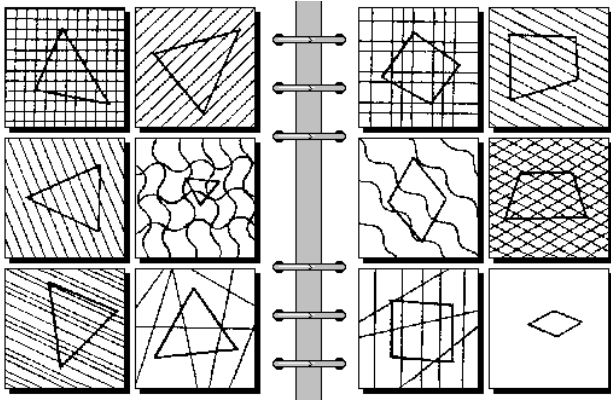


(a) Model's answer: "Left: Each image contains an even number of shapes. Right: Each image contains an odd number of shapes". Evaluated as incorrect.

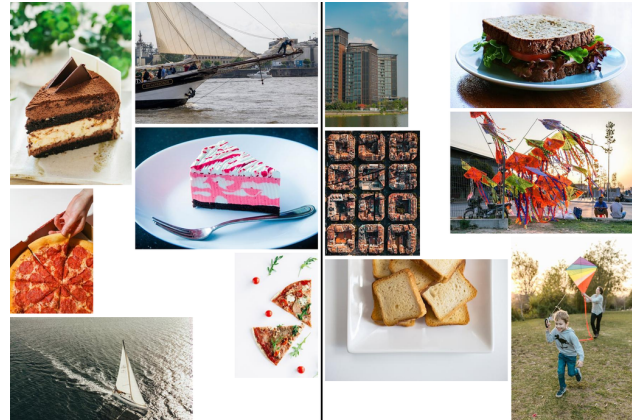


(b) Model's answer: "Left: Contains groups of three items. Right: Contains groups of five items". Evaluated as correct.

Figure 26: Synthetic BP #88 with its manually translated Bongard-RWR version. Correct answer: "Left: Three parts. Right: Five parts". Provided answers belong to GPT-4o using the Direct strategy. Overall this problem was solved by three models and four strategies.



(a) Model's answer: "Left: All images contain geometric shapes or patterns that are oriented or have components that predominantly point to the right. Right: All images contain geometric shapes or patterns that are either static (not pointing in any specific direction) or have components that do not predominantly point to the right". Evaluated as incorrect.



(b) Model's answer: "Left: All images on the left side feature objects that are primarily triangular in shape. This includes the slices of cake and pizza, which are cut into triangular pieces, and the sailboats, which have triangular sails. Right: All images on the right side feature objects or scenes that are primarily rectangular or square in shape. This includes the modern buildings, the sandwich, the grid-like urban area, the arrangement of kites, and the plate of toasted bread, all of which emphasize rectangular or square forms". Evaluated as correct.

Figure 27: Synthetic BP #98 with its manually corrected Bongard-RWR version. Correct answer: "Left: Three parts. Right: Five parts". Provided answers belong to GPT-4 Turbo using the Descriptive-direct strategy. Overall this problem was solved by two models using two different strategies.